

PURCHASE DESCRIPTION

SOCKS, ATHLETIC

This purchase description is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. **SCOPE.** This Purchase Description covers the requirements for a pair of ribbed knit, cotton/nylon/spandex athletic crew socks. The socks are intended to be worn by male and female military personnel during athletic and physical fitness activities.
2. **CLASSIFICATION.** The socks will be of one (1) type in the following classes and sizes:
 - 2.1 Classes. The socks will be available in the following classes:

Class 1 - White
Class 2 - Black 557

- 2.2 Sizes. The socks will be available in the following sizes:

SIZE SCHEDULE		
Sock Size	U.S. Men's Shoe Size	U.S. Women's Shoe Size
SMALL (S)	4 - 5.5	4 - 6.5
MEDIUM (M)	6 - 8.5	7 - 9.5
LARGE (L)	9 - 12	10 - 13
X-LARGE (XL)	12.5 - 16	

Comments, suggestions, or questions on this document should be addressed to: Department of the Army, Combat Capabilities Development Command (CCDC) Soldier Center, 10 General Greene Avenue, Natick MA 01760, ATTN: FCDD-SCD-PE.

AMSC N/A

FSC 8415

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

3. SALIENT CHARACTERISTICS.

3.1 Basic material. The basic material for the socks shall be a plain 2 x 1 rib knit leg with 1 X 1 mock rib knit top, plain jersey knit foot with terry half cushion, and a terry tuft stitch on the inside of the high heel, sole, toe and ring toe. The heel and toe of the sock shall be reinforced with nylon. The top of the finished sock shall be finished with a well-defined, non-raveling welt. The top of the sock shall have spandex laid in every course. The weight for each pair of socks shall be Small 1.8 (± 0.25) ounces, Medium 2.0 (± 0.25) ounces, Large 2.2 (± 0.25) ounces, and X-Large 2.4 (± 0.25) ounces. The material for the socks shall conform to the requirements specified in Table I, when tested as specified in Table I.

3.2 Yarn count. The crew socks shall be made with the following yarns in the specified areas of the sock (see Figure 1) when tested as specified in Table I. The foot of the sock shall be constructed with 12/1 cotton, and 2/70 nylon. The welt shall be 8/1 cotton, and spandex with two (2) ends of 1/40 nylon. The leg shall be 6/1 cotton, and spandex with two (2) ends of 1/40 nylon. The heel and toe of the sock shall be 12/1 cotton, and 2/70 nylon, and shall be reinforced with 1/70 nylon.

TABLE I. Physical requirements.

Characteristic	Requirement	Test Method
Fiber content, percent <u>1/</u>		
Cotton	75-85	AATCC TM20
Nylon	15-25	AATCC TM20
Spandex	1-4	AATCC TM20
Weight of finished pair of socks	See 3.1	<u>2/</u>
Yarn count	See 3.2	ASTM D1907
Colorfastness to:		
Laundering, after 3 cycles (min.)	3	AATCC TM61, 2A <u>3/</u>
Perspiration (min.)	4	AATCC TM15 <u>3/</u>
Crocking (min.)	3.5	AATCC TM8 <u>4/</u>
Pilling, percent (max.)	3.0	ASTM D3512
Bursting strength	Not less than 85 pounds	ASTM D3787

1/ Overall blend of cotton/ nylon and spandex must equal 100 percent

2/ Weighed on a calibrated scale

3/ Rated using the AATCC EP1, Gray Scale for Color Change.

4/ Rated using the AATCC EP8, AATCC 9-Step Chromatic Transference Scale

3.3 Color. The color of the socks shall be White for Class 1, and Black 557 for Class 2.

3.4 Sizes and measurements. The finished socks shall conform to the sizes as specified in 2.2.

3.5 Finished measurements. The socks shall conform to the finished measurements specified in Table II. The sizes Small, Medium, Large and X-Large may be abbreviated to S, M, L and XL.

TABLE II. Finished measurements (inches). 1/

SIZES	SMALL (S)	MEDIUM (M)	LARGE (L)	X-LARGE (XL)	Tolerance
Leg length A	9	10	12	13	$\pm 1/2$
Welt length B	3/4	3/4	3/4	3/4	$\pm 1/4$
Foot length C	8	9	10-1/2	11-1/2	$\pm 1/2$
Foot width D	4	4	4	4	$\pm 1/4$

(A) Leg length: From the very top of the sock to the bottom of the heel.

(B) Welt length: The double fabric area on the top of the sock.

(C) Foot length: Measure from the back of the heel to the tip of the toe.

(D) Foot width: Width from bottom of heel along the gore seam.

1/ All measurements above taken in relaxed state

3.6 Labels.

3.6.1 Size label. One (1) sock per pair shall have a size marking applied on the foot of the sock. The size may be accomplished by direct printing, stamping or stenciling. The color of the marking shall be black for Class 1 and white for Class 2.

3.6.2 Band label. All socks shall be banded in packages with an approved commercial paper band which will include NSN and bar code. The banded packages shall be three (3) pairs of socks in a pack. The contractor shall submit to the contracting officer a sample paper band for approval.

3.6.3 Identification label. Each package of banded socks shall have a commercial-type label containing the contractor's name, contract number and National Stock Number (NSN).

3.6.3.1 Barcode label. Unless otherwise specified in the contract or purchase order, each package of banded socks (3 pair) shall have a barcode label conforming to Type VIII, Class 17 of MIL-DTL-32075. The bar coding element shall be a 13 digit National Stock Number (NSN). There shall be a 12 digit Universal Product Code (UPC) number assigned for all NSN's by the contracting activity. The initials "UPC" must appear beneath the code. The bar code for NSN and UPC type shall be a medium to high density and shall be located so it is completely visible on the paper band of the socks when it is folded and/or packaged as specified. The barcodes shall be clearly legible and readable by a scanner. The label's location shall cause no damage to the item and be located on the paper band.

3.7 Washing instructions. Each pair of socks shall contain the following washing instructions, which shall be printed on the paper band:

Machine wash, warm water, with softener.
Wash with like colors. No bleach.
Tumble dry on medium heat.

3.8 Toxicity. The finished material shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 5.6. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.9 Workmanship. The socks shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels as specified in 5.4 and Table III.

4. REGULATORY REQUIREMENTS. Unless otherwise specified the offer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The socks provided shall meet the salient characteristics of this purchase description, conform to the producer's own drawings, specification, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for, under the provisions of the contract or purchase order. Reliance on contractor quality assurance systems shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract or purchase order.

5.2 Market acceptability. The socks offered must have been sold to the commercial market or the Government.

5.3 End item inspection procedures. Sampling for inspection shall be conducted in accordance with ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection of Attributes (see 7.2.3).

5.4 Visual examination. The socks shall be examined for the defects listed in Table III. The lot size shall be expressed in units of socks. The sample unit is one (1) pair of socks. The inspection level and AQL shall be as specified in the contract or purchase order. Unless otherwise specified, scoring shall be performed on defects visible from a normal inspection distance of 3-feet.

TABLE III. Visual defects.

Examine	Defect
Material and workmanship	Hole, cut, tear, missing, broken, or loose yarn, shade bar, dye streak, area of no dye penetration Any run or dropped stitches, including holes looping or seaming Knit construction of sock not as specified Any thin areas, mends, frays, ravels or raveled edges Any degree of degradation of the elastic top Noticeable variation in size (length and width) in a pair Any material defects, distorted parts or poor workmanship Bulky or uneven toe closure
Color	Color not as specified Noticeable variation in shade between two (2) socks of a pair
Cleanliness	Any spot or stain on outside clearly noticeable Any odor affecting serviceability
Labels	Omitted, incorrect, illegible, not attached where specified; barcodes, or washing instructions omitted, not readable by scanner; human-readable interpretation (HRI) omitted or illegible; barcode not visible on folded, packaged item; label/ paper band attachment causes damage to the item
Marking	Marking missing, incorrect or illegible
Packaging	Any socks not packaged in accordance with the contract or purchase order

5.5 End item dimensional examination. The socks shall be examined for conformance to the finished measurements specified in Table II. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of socks. The sample unit shall be one (1) pair of socks. The inspection level, expressed in terms of defects per hundred units, shall be as specified in the contract or purchase order.

5.6 Toxicity test. When required (see 7.6), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of the studies indicate the socks are not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 7.2.4). If the toxicity requirement (see 3.8) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required.

5.7 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.6).

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be specified in the contract or purchase order (see 7.6).

7. NOTES

7.1 Source of Government documents.

7.1.1 Copies of Government documents are available online at <https://quicksearch.dla.mil>.

7.2 Source of non-Government documents.

7.2.1 AATCC Test Methods are available online at <https://www.aatcc.org>.

7.2.2 ASTM Test Methods are available online at <https://www.astm.org>.

7.2.3 ASQ/ANSI Z1.4 – Sampling Procedures and Tables for Inspection of Attributes are available online at <https://asq.org>.

7.2.4 INFORMA HEALTHCARE - Repeat Insult Patch Test - Modified Draize Procedure - Principles and Methods of Toxicology, A Wallace Hayes (editor) are available online at <https://www.crcpress.com>.

7.3 Standard samples. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

7.4 Intended use. These socks are intended for wear by male and female military personnel during Athletic or physical fitness activities.

7.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

7.6 Ordering data. The contract or purchase order should specify the following:

- a. Title, number, and date of this purchase description.
- b. Size required (see 2.1).
- c. Labeling information (see 3.6).
- d. When toxicity testing is required (see 3.8 and 5.6).
- e. Regulatory requirements (see 4).
- f. Product conformance provisions (see 5.1).
- g. Acceptance criteria (see 5.7).
- h. Packaging requirement (see 6.1).

7.7 Subject terms.

Crew
Footwear
Physical Fitness

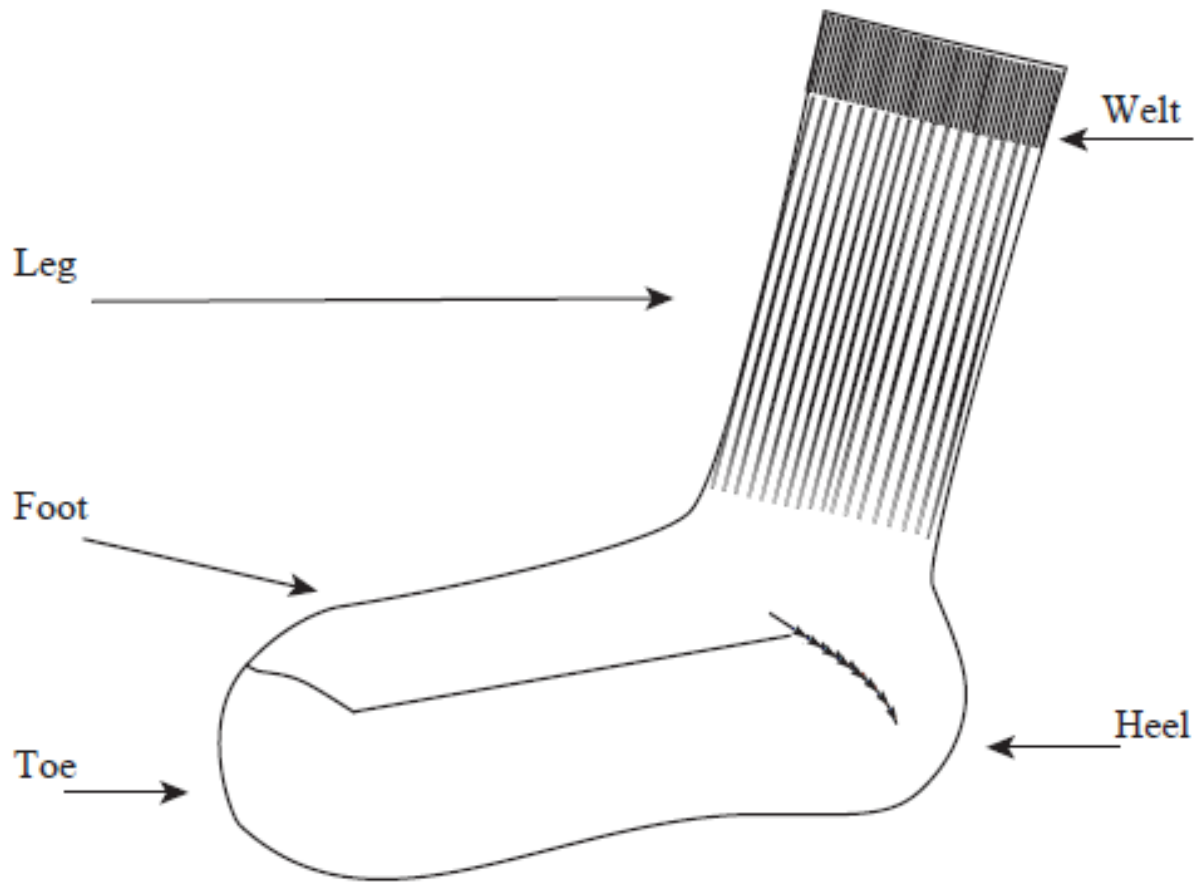


FIGURE 1. Socks, Crew

Custodian:
Army - GL

Preparing Activity:
Army - GL